
(a) What the model reads, and what it forecasts
not to scale
Slow branch: 18 eight-day means of leaf area index, soil moisture and rain → fuel amount and dryness
weather, greenness, recent fire → current conditions
Fast branch: 14 daily maps
Target
fire detected on day D + 1, D + 2 or D + 3?
D − 144 days
D − 14 days
issue day D
D + 3
time
(b) How the information is processed
1 Inputs
Fuel and moisture
leaf area index, soil moisture, rain
Weather and fire
heat, dry air (VPD), wind, NDVI, recent fire, fire intensity, fuel age
2 Encoders (ConvLSTM)
Slow memory
18 steps
Fast memory
14 steps
3 Fusion and output
Combine
slow summary added to each fast step
Risk score per 1 km pixel
darker = higher risk
Flag
pixels above one fixed threshold (chosen on 2019)